BUILD A BIN THAT THINKS.

SMART TRASH BIN

Arduino • Sensors • Servo Control • Product Design

MASKIDO INNOVATION ACADEMY

PROJECT 01

Learner Mission Workbook

Write, sketch, test and reflect as you build.

LEARNER EDITION

Name: ___________________________ Cohort/Class: ___________________________

Mission Passport

Mission	I completed it	Evidence
01 · The Challenge	☐	My prediction
02 · Meet the Team	☐	Component match
03 · Safety Gate	☐	Six promises
04 · Sensor Lab	☐	Five readings
05 · Servo Lab	☐	Two angles
06 · Connect + Code	☐	Bench test
07 · Build the Body	☐	Quality check
08 · Test + Level Up	☐	Photo + video

My mission
I will build a touch-free bin that senses a hand, opens, waits and closes safely.

01

THINK LIKE AN INVENTOR

Spot the Problem

Start with the need - not the components.

What problem does a touch-free bin solve?

____

____

____

____

____

My prediction

I think the bin will know a hand is close because...

____

____

____

____

____

Draw your first idea

Sketch here. Label the eyes, brain, muscle and lid.

02

KNOW YOUR TOOLS

Component Match

Connect each part to its job.

Component	My nickname	Its real engineering job
Arduino Uno	_______________	____
HC-SR04 sensor	_______________	____
SG90 servo	_______________	____
5V power module	_______________	____
Jumper wires	_______________	____
Foam board	_______________	____

Fast match

- The part that senses distance: ____________________________
- The part that runs the program: ____________________________
- The part that moves the lid: ____________________________
- The part that supplies stable energy: ____________________________

03

SAFETY FIRST

My Safety Promise

Engineers protect people and equipment.

- ☐ I will disconnect power before changing a wire.
- ☐ I will check 5V and GND twice.
- ☐ I will use only the approved 5V supply.
- ☐ I will ask an adult to cut or use hot glue.
- ☐ I will keep fingers away from the moving linkage.
- ☐ I will keep liquids and metal scraps away from the circuit.

Learner signature: ____________________________ Adult/instructor initials: ____________________________

Stop rule
If a wire becomes hot, the board smells unusual, or the servo buzzes continuously: switch off immediately and call the instructor.

04

SENSOR LAB

Record the Echo

Move the hand and watch the numbers change.

Trial	Approximate hand position	Sensor reading (cm)	What I noticed
1	Far	________	____________________________
2	30 cm	________	____________________________
3	20 cm	________	____________________________
4	10 cm	________	____________________________
5	5 cm	________	____________________________

My opening distance

I chose ________ cm because ____

Why did the readings change slightly?

____

____

____

____

05

SERVO LAB

Calibrate the Muscle

Find safe angles before attaching the lid.

Test	Angle	Result
Closed position	________°	____
Halfway test	________°	____
Open position	________°	____

My safe range
My lid closes at ______° and opens at ______°. The servo does not buzz or force the lid.

Draw the servo and lid linkage

Sketch the hinge, servo horn and connection to the lid.

06

CONNECT + CODE

Bench-Test Record

Test before mounting anything.

Check	Pass?	Notes
TRIG → D10 and ECHO → D9	◻	____________________________
Servo signal → D11	◻	____________________________
Servo uses approved external 5V	◻	____________________________
Arduino and servo share GND	◻	____________________________
Opens at chosen distance	◻	____________________________
Waits and closes safely	◻	____________________________

Explain the code in your own words

- The OPEN_DISTANCE_CM value controls ____
- The CLOSE_DISTANCE_CM value controls ____
- The HOLD_OPEN_MS value controls ____

07

PROTOTYPE STUDIO

Build Quality Check

Strong, square, neat and repairable.

SMART BIN BODY — CUT LIST

Recommended 5 mm foam board prototype

FRONT + BACK x2

22 × 34 cm

Measure twice. Cut once.

LEFT + RIGHT x2

22 × 34 cm

Measure twice. Cut once.

BASE x1

22 × 22 cm

Measure twice. Cut once.

TOP DECK x1

22 × 22 cm

14 × 14 cm opening

LID x1

13.5 × 13.5 cm

Measure twice. Cut once.

ELECTRONICS TRAY x1

18 × 8 cm

Measure twice. Cut once.

Quality check	Yes	Needs work
Walls stand square	☐	☐
Lid moves freely	☐	☐
Sensor openings are neat	☐	☐
Electronics tray can be removed	☐	☐
Wires stay away from waste and moving parts	☐	☐
Branding and edges are finished neatly	☐	☐

08

FINAL CHALLENGE

Test Like an Engineer

Do not stop at “it worked once.”

Cycle	Opened?	Closed?	Any problem?
1	☐	☐	________________________
2	☐	☐	________________________
3	☐	☐	________________________
4	☐	☐	________________________
5	☐	☐	________________________
6	☐	☐	________________________
7	☐	☐	________________________
8	☐	☐	________________________
9	☐	☐	________________________
10	☐	☐	________________________

The hardest problem I solved

____

____

____

My next upgrade

- ☐ Status LED
- ☐ Friendly sound
- ☐ Fill-level sensor
- ☐ My own idea: ______________________________

Submission evidence
One photo + one 30-second video + one explanation of a problem you solved.

Project title: ______________________________________ Date completed: ____________________

MISSION COMPLETE

SMART SYSTEMS BUILDER

You identified a problem, built a working system, tested it repeatedly and improved it. That is engineering.

Badge approval
Award after the final evidence is reviewed and the learner earns at least 14/20 with safety and function both satisfactory.

My engineer reflection

The part I am proudest of is ____

The next version of my invention will ____

Learner name: _____________________________________ Final score: ______ / 20

Instructor sign-off: ________________________________ Date: ____________________________